

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801H - MODERN CONSTRUCTION MATERIALS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which type of atomic bonding involves sharing of electrons between atoms?
 - A) Ionic bonding
 - B) Metallic bonding
 - C) Covalent bonding
 - D) Hydrogen bonding
 - 2 In crystalline solids, atoms are arranged in:
 - A) Random pattern
 - B) Short-range order only
 - C) Long-range periodic order
 - D) Amorphous structure
 - 3 Diffusion in solids primarily occurs due to:
 - A) External load
 - B) Temperature gradient
 - C) Atomic movement driven by concentration gradient
 - D) Grain boundary strengthening
 - 4 Stress is defined as:
 - A) Deformation per unit length
 - B) Force per unit area
 - C) Resistance to deformation
 - D) Energy stored in material
 - 5 According to Griffith's theory, brittle fracture occurs when:
 - A) Yield stress is exceeded
 - B) Elastic limit is crossed
 - C) Energy released equals surface energy
 - D) Plastic deformation dominates
 - 6 Rheology is the study of:
 - A) Heat transfer in materials
 - B) Elastic behaviour only
 - C) Flow and deformation of materials
 - D) Chemical durability

- 7 Which of the following is a thermoplastic polymer?
 - A) Bakelite
 - B) Epoxy resin
 - C) PVC
 - D) Polyester resin

- 8 The primary advantage of Fibre Reinforced Polymers (FRP) is:
 - A) High density
 - B) Low corrosion resistance
 - C) High strength-to-weight ratio
 - D) Brittleness

- 9 Bitumen is best described as a:
 - A) Brittle material
 - B) Thermosetting material
 - C) Viscoelastic material
 - D) Crystalline solid

- 10 Which component of concrete primarily contributes to strength?
 - A) Fine aggregate
 - B) Coarse aggregate
 - C) Cement paste
 - D) Water

- 11 Glass is classified as a:
 - A) Crystalline solid
 - B) Semi-crystalline solid
 - C) Amorphous solid
 - D) Polymeric solid

- 12 Polymer-based waterproofing materials are mainly used to:
 - A) Increase compressive strength
 - B) Prevent water ingress
 - C) Improve thermal conductivity
 - D) Reduce dead load

Q2. Solve the following.

- A) Explain the scope of Science, Engineering, and Technology of Materials. Discuss the importance of atomic bonding in determining material properties. 6
- B) Describe different types of atomic bonding and explain the structure of solids. Differentiate between crystalline and amorphous solids. 6

Q3. Solve the following.

- A) Explain the movement of atoms in solids. Discuss how atomic movement contributes to the development of microstructure in engineering materials. 6
- B) Discuss surface properties of materials and explain the response of materials to applied stress with reference to stress-strain behaviour. 6

Q4. Solve Any Two of the following.

- A) Explain different failure theories used for engineering materials. Discuss their applicability to ductile and brittle materials. 6
- B) Define fracture mechanics and explain the concepts of stress concentration, crack propagation, and fracture toughness. 6
- C) Discuss rheological and thermal properties of construction materials. Explain the criteria for selection of wood and wood products in engineering applications. 6

Q5. Solve Any Two of the following.

- A) Discuss the properties, defects, and uses of wood and wood products. Explain the classification and engineering applications of polymers. 6
- B) Explain Fibre Reinforced Polymers (FRP) and discuss their advantages and applications in construction. Also describe the basic structure and properties of metals. 6
- C) Discuss advanced properties of metals and explain the composition, properties, and engineering uses of bituminous materials. 6

Q6. Solve Any Two of the following.

- A) Explain the constituents of concrete and discuss the fresh and hardened properties of concrete. 6
- B) Discuss special concretes and explain the properties and applications of glass as a construction material. 6
- C) Explain different waterproofing materials and polymer floor finishes. Discuss the role and types of anchors used in construction. 6

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